

LOGIC BUILDING X Guess The Output?

(BEFORE STARTING DSA)

Phase 1 – Warm-Up: Thinking Like a Compiler

Phase 2 – Variables, Data Types & Literals

Phase 3 – Operators Mastery

Phase 4 – Conditional Execution (If / Else)

Phase 5 – Loops

Phase 6 – Variables Scope & Lifetime

Phase 7 – Arrays

Phase 8 – Strings

Phase 9 – Type Casting & Promotions

Phase 10 – Mixed Logic Challenges

 **Goal:** Strong your language logics

Topics covered: Complete Language fundamentals [Think like Java compiler]

Target Questions: 400+

Phase 1 – Warm -Up : Thinking Like a Compiler

Q1

```
int a = 5;  
System.out.println(a);
```

Output: 5

Reason: a store 5, print displays its value.

Q2

```
System.out.println(10);
```

Output: 10

Reason: Literal value is printed directly.

Q3

```
int x = 3;  
int y = 4;  
System.out.println(x + y);
```

Output: 7

Reason: Expression evaluated first, then printed.

Q4

```
int a = 10;  
System.out.println(a + 5);
```

Output: 15

Reason: a is 10, expression becomes 10 + 5.

Takeaway:

Variables substitute values before calculation.

Q5

```
int a = 5;  
a = 8;  
System.out.println(a);
```

Output: 8

Reason: The latest assignment overwrites old values.

Q6

```
int x;  
x = 20;  
System.out.println(x);
```

Output:

20

Reason:

Assignment happens before print.

Takeaway:

Declaration ≠ assignment.

Q7

```
int a = 2;  
System.out.println(a * 3);
```

Output:

6

Reason:

Multiplication evaluated before print.

Takeaway:

Arithmetic executes before output.

Q8

```
int a = 9;  
System.out.println(a / 3);
```

Output: 3

Reason: Integer division with no remainder.

Takeaway:

The division result depends on data type (int here).

Q9

```
int a = 5;  
System.out.println(a + 2);  
System.out.println(a);
```

Output:

7

5

Reason: $a + 2$ does not change a .

Takeaway:

Expressions do not modify variables unless assigned.

Q10

```
int a = 4;  
a = a + 3;  
System.out.println(a);
```

Output:

7

Reason:

Old value used on RHS, new value stored.

Takeaway:

Assignment updates variable memory.

Q11

```
int x = 10;  
int y = x;  
x = 20;  
System.out.println(y);
```

Output: 10

Reason:

y stores a copy of x's value, not a link.

Takeaway:

Primitive assignment copies values.

Q12

```
int a = 2;  
int b = 3;  
int c = a + b;  
System.out.println(c);
```

Output:

5

Reason:

Expression evaluated, result stored in c.

Takeaway:

Variables can store expression results.

Q13

```
int a = 5;  
System.out.println(a + a);
```

Output:

10

Reason:

Same variable value used twice.

Takeaway:

Each variable reference uses current value.

Q14

```
int a = 1;  
int b = 2;  
a = b;  
System.out.println(a);
```

Output:

2

Reason:

a is overwritten by b's value.

Takeaway:

Assignment replaces previous data.

Q15

```
int a = 10;  
int b = 5;  
System.out.println(a - b);
```

Output:

5

Reason:

Subtraction evaluated before print.

Takeaway:

Order: evaluate → print.

Q16

```
int a = 3;  
System.out.println(a * a);
```

Output:

9

Reason:

Multiplication uses same value twice.

Takeaway:

Variables can appear multiple times in an expression.

Q17

```
int a = 5;  
System.out.println(a + 1);  
a = 10;  
System.out.println(a + 1);
```

Output:

6

11

Reason:

First print uses a = 5, second uses updated value.

Takeaway:

Execution order matters more than code position.

Q18

```
int a = 2;  
int b = a + 3;  
a = 10;  
System.out.println(b);
```

Output:

5

Reason:

b is computed before a changes.

Takeaway:

Variables do not auto-update past computations.

Q19

```
int a = 1;  
System.out.println(a);  
a = a + 1;  
System.out.println(a);
```

Output:

1

2

Reason:

Print → update → print.

Takeaway:

Read code top to bottom, not all at once.

Q20

```
int a = 5;  
int b = 10;  
System.out.println(a);  
a = b;  
System.out.println(a);
```

Output:

5

10

Phase 2 – Variables, Data Types & Literals

Q21

```
int a = 10;  
System.out.println(a);
```

Output: 10

Takeaway:

int stores whole numbers.

Q22

```
double d = 5;  
System.out.println(d);
```

Output: 5.0

Reason:

Integer literal stored in a double becomes decimal.

Takeaway:

Wider types can store smaller values.

Q23

```
char c = 'A';  
System.out.println(c);
```

Output: A

Takeaway:

char stores a single character in single quotes.

Q24

```
boolean b = true;  
System.out.println(b);
```

Output: true

Takeaway:

Boolean has only two values: true / false.

Q25

```
float f = 2.5f;  
System.out.println(f);
```

Output:

2.5

Takeaway:

Float literals require f.

Q26

```
long x = 100;  
System.out.println(x);
```

Output:

100

Takeaway:

Integer literals default to int but fit in long.

Q27

```
int a = 5;  
a = 7;  
System.out.println(a);
```

Output:

7

Takeaway:

Reassignment overwrites old value.

Q28

```
double d = 3.14;  
System.out.println(d);
```

Output:

3.14

Takeaway:

Decimal literals are double by default.

Q29

```
char c = '7';  
System.out.println(c);
```

Output:

7

Takeaway:

Character '7' is not number 7.

Q30

```
boolean flag = false;  
System.out.println(flag);
```

Output:

false

Takeaway:

Boolean prints exactly its value.

Q31

```
int a = 10;  
double b = a;  
System.out.println(b);
```

Output:

10.0

Reason:

Implicit widening conversion.

Takeaway:

Smaller types fit into larger types automatically.

Q32

```
char c = 'A';  
int x = c;  
System.out.println(x);
```

Output:

65

Reason:

Characters are stored as numeric codes.

Takeaway:

char is internally a number.

Q33

```
int a = 5;  
double b = 2.0;  
System.out.println(a + b);
```

Output:

7.0

Reason:

Expression promoted to double.

Takeaway:

Mixed-type expressions upgrade to wider type.

Q34

```
long l = 10;  
int x = l;  
System.out.println(x);
```

Output:

 Compilation Error

Reason:

Narrowing is not allowed without casting.

Takeaway:

Java prevents data loss by default.

Q35

```
float f = 10;  
System.out.println(f);
```

Output:

10.0

Reason:

Int literal fits into float.

Q36

```
char c = 66;  
System.out.println(c);
```

Output:

B

Reason:

66 maps to character 'B'.

Takeaway:

Chars can be assigned numeric values.

Q37

```
int a = 5;  
int b = a;  
a = 10;  
System.out.println(b);
```

Output:

5

Takeaway:

Primitive assignments copy values, not references.

Q38

```
double d = 5 / 2;  
System.out.println(d);
```

Output:

2.0

Reason:

Integer division happens first.

Takeaway:

Expression type depends on operands.

Q39

```
float f = 3.5;  
System.out.println(f);
```

Output:

✗ Compilation Error

Reason:

Decimal literal defaults to double.

Takeaway:

Literal type matters.

Q40

```
char c = 'a';  
System.out.println(c + 1);
```

Output:

98

Reason:

'a' = 97, arithmetic converts to int.

Takeaway:

Arithmetic on char produces int.

Q41

```
long l = 100000000000;  
System.out.println(l);
```

Output:

✗ Compilation Error

Reason:

Integer literal too large for int.

Takeaway:

Large literals need L.

Q42

```
float f = 2.5;  
System.out.println(f);
```

Output:

✗ Compilation Error

Reason:

Double literal cannot fit into float implicitly.

Takeaway:

Java is strict with decimals.

Q43

```
char c = 'A';  
System.out.println(c + c);
```

Output:

130

Reason:

65 + 65 = 130

Takeaway:

Chars convert to numbers in expressions.

Q44

```
boolean b = true;  
System.out.println(b + 1);
```

Output:

✗ Compilation Error

Reason:

Boolean cannot participate in arithmetic.

Takeaway:

Boolean is not numeric in Java.

Q45

```
double d = 10 / 3;  
System.out.println(d);
```

Output:

3.0

Reason:

Integer division happens before assignment.

Phase 3 – Operators Mastery

Q46

```
int a = 5;  
System.out.println(a + 2);
```

Output: 7

Takeaway: Simple arithmetic evaluation.

Q47

```
int a = 10;  
System.out.println(a - 3);
```

Output: 7

Q48

```
int a = 4;  
System.out.println(a * 2);
```

Output: 8

Q49

```
int a = 9;  
System.out.println(a / 3);
```

Output: 3

Q50

```
int a = 10;  
System.out.println(a % 3);
```


Output: 1
Takeaway: Remainder operator.

Q51

```
int a = 5;  
System.out.println(+a);
```

Output: 5

Q52

```
int a = 5;  
System.out.println(-a);
```

Output: -5

Q53

```
int a = 5;  
a += 3;  
System.out.println(a);
```

Output: 8

Q54

```
int a = 10;  
a -= 4;  
System.out.println(a);
```

Output: 6

Q55

```
int a = 2;  
a *= 3;  
System.out.println(a);
```

Output: 6

Q56

```
int a = 5;  
System.out.println(a > 3);
```

Output: true

Q57

```
int a = 5;  
System.out.println(a < 3);
```

Output: false

Q58

```
int a = 5;  
System.out.println(a == 5);
```

Output: true

Q59

```
int a = 5;  
System.out.println(a != 5);
```

Output: false

Q60

```
boolean x = true;  
boolean y = false;  
System.out.println(x && y);
```

Output: false

Q61

```
boolean x = true;  
boolean y = false;  
System.out.println(x || y);
```

Output: true

Q62

```
boolean x = true;  
System.out.println(!x);
```

Output: false

Q63

```
int a = 5;  
System.out.println(a + 2 * 3);
```

Output: 11
Takeaway: * before.

Q64

```
int a = 10;  
System.out.println(a / 2 + 3);
```

Output: 8

Q65

```
int a = 10;  
System.out.println((a + 2) * 3);
```

Output: 36

Q66

```
int a = 5;  
System.out.println(a++);
```

Output: 5
Takeaway: Post-increment returns old value.

Q67

```
int a = 5;  
System.out.println(++a);
```

Output: 6
Takeaway: Pre-increment increments first.

Q68

```
int a = 5;  
a++;  
System.out.println(a);
```

Output: 6

Q69

```
int a = 5;  
--a;  
System.out.println(a);
```

Output: 4

Q70

```
int a = 5;  
System.out.println(a--);
```

Output: 5

Q71

```
int a = 5;  
System.out.println(--a);
```

Output: 4

Q72

```
int a = 5;  
System.out.println(a + ++a);
```

Output: 11

Q73

```
int a = 5;  
System.out.println(a + a++);
```

Output: 10

Q74

```
int a = 5;  
int b = a++;  
System.out.println(a + " " + b);
```

Output: 6 5

Q75

```
int a = 5;  
int b = ++a;  
System.out.println(a + " " + b);
```

Output: 6 6

Q76

```
int a = 5;  
System.out.println(a++ + ++a);
```

Output: 12

Q77

```
int a = 10;  
System.out.println(a-- - --a);
```

Output: 2

Q78

```
int a = 3;  
int b = 4;  
System.out.println(a > b && ++a > 0);
```

Output: false

Takeaway: Short-circuit stops increment.

Q79

```
int a = 3;  
int b = 4;  
System.out.println(a < b || ++a > 0);
```

Output: true

Q80

```
int a = 5;
```

```
System.out.println(a > 2 && a++ > 5);
```

Output: false

Q81

```
int a = 5;  
System.out.println(a++ > 5 || a > 5);
```

Output: true

Q82

```
int a = 5;  
int b = 10;  
System.out.println(a + b * 2);
```

Output: 25

Q83

```
int a = 10;  
System.out.println(a / 2 * 3);
```

Output: 15

Q84

```
int a = 5;  
System.out.println(a++ + a++ + a);
```

Output: 18

Q85

```
int a = 5;  
System.out.println(++a + ++a);
```

Output: 13

Q86

```
int a = 10;  
int b = 5;  
System.out.println(a > b && b++ > 0);
```

Output: true

Q87

```
int a = 0;  
System.out.println(a++ == 0 || ++a == 2);
```

Output: true

Q88

```
int a = 1;  
System.out.println(++a == 2 && a++ == 2);
```

Output: true

Q89

```
int a = 5;  
System.out.println(a-- + --a);
```

Output: 8

Q90

```
int a = 5;  
System.out.println(a++ * 2);
```


Output: 10

Q91

```
int a = 5;  
System.out.println(++a * 2);
```

Output: 12

Q92

```
int a = 5;  
System.out.println(a += 3);
```

Output: 8

Q93

```
int a = 5;  
System.out.println(a *= 2 + 3);
```

Output: 25

Q94

```
int a = 10;  
System.out.println(a /= 3);
```

Output: 3

Q95

```
int a = 5;  
System.out.println(a += a++ + ++a);
```

Output: 18

Q96

```
int a = 2;  
System.out.println(a++ + ++a * a);
```

Output: 18

Q97

```
int a = 3;  
System.out.println(a++ + a++ + ++a);
```

Output: 13

Q98

```
int a = 1;  
System.out.println(a++ + ++a + a++);
```

Output: 7

Q99

```
int a = 5;  
System.out.println(a > 3 && a++ > 6 || a > 4);
```

Output: true

Q100

```
int a = 5;  
System.out.println(a < 3 || ++a > 5 && a > 6);
```

Output: false

Q101

```
int a = 5;  
System.out.println(a++ + ++a + ++a);
```

Output: 20

Q102

```
int a = 2;  
System.out.println(++a + a++ * ++a);
```

Output: 18

Q103

```
int a = 3;  
System.out.println(a++ > 3 && ++a > 5 || a++ > 3);
```

Output: true

Q104

```
int a = 0;  
System.out.println(a++ == 0 && ++a == 2 || a++ == 2);
```

Output: true

Q105

```
int a = 4;  
System.out.println(a++ + a++ * --a);
```

Output: 29

Q106

```
int a = 5;
```

```
System.out.println(a-- + a++ * ++a);
```

Output: 29

Q107

```
int a = 1;  
System.out.println(++a + a++ + a++ + ++a);
```

Output: 12

Q108

```
int a = 2;  
System.out.println(a++ * ++a + a++);
```

Output: 12

Q109

```
int a = 3;  
System.out.println(a++ + ++a * a++ + a);
```

Output: 34

Q110

```
int a = 5;  
System.out.println(a > 3 || ++a > 6 && a++ > 5);
```

Output: true

Q111

```
int a = 5;  
System.out.println(a++ > 5 && ++a > 6 || ++a > 6);
```

Output: true

Q112

```
int a = 2;  
System.out.println(a++ + ++a * a++ + ++a);
```

Output: 24

Q113

```
int a = 1;  
System.out.println(a++ + ++a * ++a + a++);
```

Output: 17

Q114

```
int a = 3;  
System.out.println(a++ + a++ + ++a * a);
```

Output: 43

Q115

```
int a = 1;  
System.out.println(a++ + a++ + ++a + a);
```

Output: 11

Q116

```
int a = 1, b = 2;  
System.out.println(a++ + ++b + a++ + b++);
```

Output: 6

Q117

```
int x = 2, y = 3;  
System.out.println(++x + y++ + ++x + y);
```

Output: 11

Q118

```
int a = 3, b = 4;  
System.out.println(a++ + b++ * ++a + b);
```

Output: 28

Q119

```
int x = 4, y = 2;  
System.out.println(++x * y++ + x++ + ++y);
```

Output: 19

Q120

```
int a = 1, b = 2;  
System.out.println(a++ + ++b * ++a + b++);
```

Output: 13

Q121

```
int x = 2, y = 3;  
System.out.println(x++ * ++y + x++ + ++y);
```

Output: 16

Q122

```
int a = 3, b = 1;  
System.out.println(a++ + ++a * b++ + b);
```

Output: 10

Q123

```
int x = 5, y = 3;  
System.out.println(x-- + --y * x-- + y);
```

Output: 15

Q124

```
int a = 4, b = 2;  
System.out.println(--a + a-- * --b + b);
```

Output: 7

Q125

```
int x = 2, y = 3;  
System.out.println(++x + y++ * x++ + ++y);
```

Output: 17

Q126

```
int a = 1, b = 2;  
System.out.println(a++ + b++ + ++a * b++ + a);
```

Output: 15

Q127

```
int x = 3, y = 2;  
System.out.println(++x * y++ + x++ + y);
```

Output: 15

Q128

```
int a = 2, b = 3;  
System.out.println(a++ + ++b + a++ * ++b);
```

Output: 21

Q129

```
int x = 4, y = 1;  
System.out.println(x++ * ++y + ++x + y++);
```

Output: 16

Q130

```
int a = 3, b = 2;  
System.out.println(++a + b++ + ++a * ++b);
```

Output: 26

Phase 4 – Conditional Execution (If / Else)

Q131

```
int a = 5;  
if (a > 3)  
    System.out.println("A");
```

Output: A

Q132

```
int a = 2;  
if (a > 3)  
    System.out.println("A");  
else  
    System.out.println("B");
```

Output: B

Q133

```
int a = 10;  
if (a == 10)  
    System.out.println("YES");
```

Output: YES

Q134

```
int a = 5;  
if (a != 5)  
    System.out.println("X");  
else  
    System.out.println("Y");
```

Output: Y

Q135

```
int a = 7;  
if (a > 5 && a < 10)  
    System.out.println("OK");
```

Output: OK

Q136

```
int a = 4;  
if (a > 5 || a < 3)  
    System.out.println("YES");  
else  
    System.out.println("NO");
```

Output: NO

Q137

```
int a = 5;  
if (a > 2)  
    if (a > 4)  
        System.out.println("A");
```

Output: A

Q138

```
int a = 1;  
if (a > 0)  
    if (a > 2)  
        System.out.println("X");  
    else  
        System.out.println("Y");
```

Output: Y

Q139

```
int a = 0;  
if (a == 0)  
    System.out.println("ZERO");
```

Output: ZERO

Q140

```
int a = 3;  
if (a > 5)  
    System.out.println("A");  
else if (a > 2)  
    System.out.println("B");
```

Output: B

Q141

```
int a = 10;  
if (a < 5)  
    System.out.println("X");  
else if (a < 8)  
    System.out.println("Y");  
else  
    System.out.println("Z");
```

Output: Z

Q142

```
int a = 5;  
if (true)  
    System.out.println("A");  
else  
    System.out.println("B");
```

Output: A

Q143

```
int a = 5;  
if (false)  
    System.out.println("A");  
System.out.println("B");
```

Output:

B

Q144

```
int a = 3;  
if (a > 2)  
    System.out.println("A");  
System.out.println("B");
```

Output:

A
B

Q145

```
int a = 2;  
if (a > 3);  
    System.out.println("X");
```

Output:

X

Q146

```
int a = 5;  
if (a > 3)  
    if (a < 4)  
        System.out.println("A");  
    else  
        System.out.println("B");
```

Output: B

Q147

```
int a = 2;
if (a > 3) {
    if (a > 1)
        System.out.println("A");
    else
        System.out.println("B");
}
```

Output: nothing will get printed

Q148

```
int a = 5;
if (a > 3)
    System.out.println("A");
    System.out.println("B");
```

Output:

A
B

Q149

```
int a = 0;
if (a++ > 0)
    System.out.println("A");
else
    System.out.println(a);
```

Output: 1

Q150

```
int a = 1;
```

```
if (++a > 2)
    System.out.println("X");
else
    System.out.println(a);
```

Output: 2

Q151

```
int a = 5;
if (a > 3 && a++ > 5)
    System.out.println("A");
else
    System.out.println(a);
```

Output: 6

Q152

```
int a = 5;
if (a < 3 || ++a > 5)
    System.out.println(a);
```

Output: 6

Q153

```
int a = 0;
if (a == 0)
    if (a++ == 0)
        System.out.println("A");
    else
        System.out.println("B");
```

Output: A

Q154

```
int a = 1;
```

```
if (a > 0)
    if (a > 2)
        System.out.println("X");
    else
        if (a == 1)
            System.out.println("Y");
```

Output: Y

Q155

```
int a = 5;
if (a > 10)
    System.out.println("X");
else if (a > 4)
    System.out.println("Y");
else
    System.out.println("Z");
```

Output: Y

Q156

```
int a = 1;
if (a > 2){
    if (a < 2)
        System.out.println("X");
    else
        System.out.println("Y");
}
```

Output: *(No Output)*

Q157

```
int a = 5;
if (a > 3)
    if (a > 6)
        System.out.println("X");
    else if (a > 4)
        System.out.println("Y");
```

Output: Y

Q158

```
int a = 1;
if (a++ == 1 && ++a == 3)
    System.out.println("YES");
else
    System.out.println(a);
```

Output: YES

Q159

```
int a = 2;
if (++a == 3 || a++ == 3)
    System.out.println(a);
```

Output: 3

Q160

```
int a = 5;
if (a > 3)
    if (a > 4)
        if (a > 6)
            System.out.println("X");
        else
            System.out.println("Y");
```

Output: Y

Q161

```
int a = 0;
if (a++ == 1)
    System.out.println("X");
else if (++a == 2)
```



```
System.out.println("Y");
```

Output: Y

Q162

```
int a = 5;  
if (a > 4)  
    if (a > 6)  
        System.out.println("X");  
else  
    System.out.println("Y");
```

Output: *(No Output)*

Q163

```
int a = 3;  
if (a == 3)  
    if (a > 2)  
        System.out.println("A");  
else  
    System.out.println("B");
```

Output: A

Q164

```
int a = 1;  
if (a > 0)  
    if (a > 1)  
        System.out.println("X");  
    else  
        System.out.println("Y");  
else  
    System.out.println("Z");
```

Output: Y

Q165

```
int a = 2;  
if (a > 1)  
    if (a > 3)  
        System.out.println("X");  
    else  
        System.out.println("Y");
```

Output: Y

Q166

```
int a = 1;  
if (a++ == 1)  
    if (++a == 3)  
        System.out.println("A");  
    else  
        System.out.println("B");
```

Output: B

Q167

```
int a = 0;  
if (false)  
    System.out.println("X");  
else if (true)  
    if (a++ == 0)  
        System.out.println("Y");
```

Output: Y

Q168

```
int a = 5;  
if (a > 3)  
    if (a > 10)  
        System.out.println("X");
```

```
else
    System.out.println("Y");
```

Output: *(No Output)*

Q169

```
int a = 2;
if (a++ == 2)
    if (a++ == 3)
        System.out.println("A");
    else
        System.out.println("B");
```

Output: B

Q170

```
int a = 3;
if (a > 2)
    if (a++ == 3)
        if (++a == 5)
            System.out.println("X");
        else
            System.out.println("Y");
```

Output: X

Q171

```
int a = 1;
if (a > 2)
    System.out.println("X");
else if (a++ == 1)
    if (a == 2)
        System.out.println("Y");
```

Output: Y

Q172

```
int a = 5;  
if (a++ > 5)  
    System.out.println("X");  
else if (++a > 6)  
    System.out.println("Y");
```

Output: Y

Q173

```
int a = 0;  
if (a++ == 1)  
    System.out.println("X");  
else if (a++ == 1)  
    System.out.println("Y");
```

Output: Y

Q174

```
int a = 2;  
if (++a == 3)  
    if (a++ == 3)  
        System.out.println("X");  
    else  
        System.out.println("Y");
```

Output: X

Q175

```
int a = 1;  
if (a++ == 1)  
    if (a++ == 2)  
        if (++a == 4)  
            System.out.println("PASS");  
    else  
        System.out.println("FAIL");
```

Output: PASS

Phase 5 – Loops

Q176

```
for (int i = 1; i <= 5; i++)  
    System.out.print(i);
```

Output: 12345

Q177

```
for (int i = 5; i >= 1; i--)  
    System.out.print(i);
```

Output: 54321

Q178

```
int i = 1;  
while (i <= 3) {  
    System.out.print(i);  
    i++;  
}
```

Output: 123

Q179

```
int i = 3;  
while (i > 0)  
    System.out.print(i--);
```

Output: 321

Q180

```
int i = 1;
do {
    System.out.print(i);
    i++;
} while (i <= 3);
```

Output: 123

Q181

```
int i = 1;
do {
    System.out.print(i);
} while (i++ < 3);
```

Output: 123

Q182

```
for (int i = 0; i < 5; i += 2)
    System.out.print(i);
```

Output: 024

Q183

```
for (int i = 1; i <= 5; i++)
    if (i % 2 == 0)
        System.out.print(i);
```

Output: 24

Q184

```
int i = 0;
while (i++ < 3)
    System.out.print(i);
```

Output: 123

Q185

```
for (int i = 1; i < 5; i++)  
    System.out.print(i * 2);
```

Output: 2468

Q186

```
int i = 1;  
while (true) {  
    if (i == 3)  
        break;  
    System.out.print(i++);  
}
```

Output: 12

Q187

```
for (int i = 1; i <= 5; i++) {  
    if (i == 3)  
        continue;  
    System.out.print(i);  
}
```

Output: 1245

Q188

```
for (int i = 0; i < 3; i++)  
    System.out.print(i);  
System.out.print("X");
```

Output: 012X

Q189

```
int i = 5;
while (--i > 0)
    System.out.print(i);
```

Output: 4321

Q190

```
for (int i = 1; i <= 3; i++)
    for (int j = 1; j <= 2; j++)
        System.out.print(i);
```

Output: 112233

Q191

```
for (int i = 1; i <= 3; i++)
    for (int j = 1; j <= i; j++)
        System.out.print(j);
```

Output: 112123

Q192

```
int i = 0;
while (i < 3)
    System.out.print(++i);
```

Output: 123

Q193

```
for (int i = 1; i <= 5; i++) {
    if (i > 3)
        break;
    System.out.print(i);
}
```

Output: 123

Q194

```
int i = 0;
do {
    System.out.print(i);
    i++;
} while (i < 0);
```

Output: 0

Q195

```
for (int i = 0; i < 3; i++)
    System.out.print(i++);
```

Output: 02

Q196

```
for (int i = 0; i < 5; i++) {
    System.out.print(i);
    i++;
}
```

Output: 024

Q197

```
int i = 0;
while (i < 5) {
    System.out.print(i);
    i += 2;
}
```

Output: 024

Q198

```
for (int i = 1; i <= 3; i++)  
    for (int j = i; j <= 3; j++)  
        System.out.print(j);
```

Output: 123233

Q199

```
int i = 1;  
while (i <= 5) {  
    if (i == 3) {  
        i++;  
        continue;  
    }  
    System.out.print(i);  
    i++;  
}
```

Output: 1245

Q200

```
for (int i = 1; i <= 5; i++) {  
    if (i == 4)  
        break;  
    System.out.print(i);  
}
```

Output: 123

Q201

```
int i = 0;  
for (; i < 3; ) {  
    System.out.print(i);  
    i++;  
}
```

Output: 012

Q202

```
for (int i = 1; i <= 3; i++)  
    for (int j = 1; j <= 3; j++)  
        if (i == j)  
            System.out.print(i);
```

Output: 123

Q203

```
int i = 0;  
while (i < 5)  
    System.out.print(i++);
```

Output: 01234

Q204

```
for (int i = 3; i-- > 0;)  
    System.out.print(i);
```

Output: 210

Q205

```
int i = 1;  
do {  
    System.out.print(i++);  
} while (i < 1);
```

Output: 1

Q206

```
int i = 1;  
while (i <= 3) {  
    System.out.print(i);
```

```
    i += i;  
}
```

Output: 13

Q207

```
for (int i = 0; i < 3; i++)  
    for (int j = 0; j < i; j++)  
        System.out.print(j);
```

Output: 01

Q208

```
int i = 5;  
while (i-- > 0)  
    System.out.print(i);
```

Output: 43210

Q209

```
for (int i = 0; i < 3; i++) {  
    System.out.print(i);  
    for (int j = 0; j < 2; j++)  
        break;  
}
```

Output: 012

Q210

```
int i = 0;  
while (true) {  
    if (i == 2)  
        break;  
    System.out.print(i++);  
}
```

Output: 01

Q211

```
for (int i = 1; i <= 4; i++) {  
    if (i == 2)  
        continue;  
    System.out.print(i);  
}
```

Output: 134

Q212

```
for (int i = 1; i <= 3; i++)  
    for (int j = 3; j >= i; j--)  
        System.out.print(j);
```

Output: 321323

Q213

```
int i = 0;  
do {  
    i++;  
    System.out.print(i);  
} while (i < 3);
```

Output: 123

Q214

```
for (int i = 1; i <= 3; i++)  
    for (int j = 1; j <= 3; j++)  
        if (i + j == 4)  
            System.out.print(i);
```

Output: 123

Q215

```
int i = 10;
while (i > 5)
    System.out.print(--i);
```

Output: 9876

Q216

```
for (int i = 0; i < 3; i++)
    System.out.print(++i);
```

Output: 13

Q217

```
int i = 1;
while (i < 10) {
    i *= 2;
    System.out.print(i);
}
```

Output: 24816

Q218

```
for (int i = 5; i > 0; i /= 2)
    System.out.print(i);
```

Output: 521

Q219

```
int i = 0;
while (i < 3)
    System.out.print(i = i + 1);
```

Output: 123

Q220

```
for (int i = 1; i <= 3; i++)  
    System.out.print(i + (i++));
```

Output: 26

Q221

```
for (int i = 0; i < 3; i++)  
    System.out.print(i = i++);
```

Output: 012

Q222

```
int i = 0;  
while (i < 3)  
    System.out.print(i = i++);
```

Output: 000

Q223

```
for (int i = 1; i <= 3; i++)  
    System.out.print(i++ + ++i);
```

Output: 4

Q224

```
int i = 1;  
while (i++ <= 3)  
    System.out.print(++i);
```

Output: 35

Q225

```
for (int i = 1; i <= 3; i++)  
    for (int j = 1; j <= 3; j++)  
        if (i == j)  
            break;  
        else  
            System.out.print(j);
```

Output: 112

Q226

```
int i = 0;  
for (; i < 3; i++)  
    System.out.print(i++);
```

Output: 02

Q227

```
for (int i = 1; i <= 3; i++) {  
    for (int j = 1; j <= 3; j++) {  
        if (j == 2)  
            continue;  
        System.out.print(j);  
    }  
}
```

Output: 131313

Q228

```
int i = 1;  
while (i <= 3) {  
    System.out.print(i);  
    i = i * i;  
}
```


Output: 11111111111111... (infinite loop)

Q229

```
for (int i = 0; i < 3; i++)  
    System.out.print(i + ++i);
```

Output: 13

Q230

```
int i = 3;  
while (i-- > 0)  
    System.out.print(--i);
```

Output: 10-1

Q231

```
for (int i = 0; i < 5; i++) {  
    if (i == 3)  
        continue;  
    System.out.print(i);  
}
```

Output: 0124

Q232

```
int i = 5;  
while (i > 0) {  
    System.out.print(i);  
    i /= 2;  
}
```

Output: 521

Q233

```
for (int i = 1; i <= 3; i++)  
    for (int j = i; j <= 3; j++)  
        if (j == 2)  
            break;  
        else  
            System.out.print(j);
```

Output: 113

Q234

```
int i = 0;  
while (++i < 3)  
    System.out.print(i++);
```

Output: 12

Q235

```
for (int i = 1; i <= 3; i++)  
    System.out.print(++i + i++);
```

Output: 4

Q236

```
int i = 1;  
do {  
    System.out.print(i++);  
} while (++i <= 3);
```

Output: 13

Q237

```
for (int i = 0; i < 3; i++)  
    System.out.print(i + i++);
```

Output: 13

Q238

```
int i = 1;
while (i < 5)
    System.out.print(i = i++ + 1);
```

Output: 2345

Q239

```
for (int i = 1; i <= 3; i++)
    for (int j = 1; j <= 3; j++)
        if (i == 2)
            break;
        else
            System.out.print(j);
```

Output: 123123

Q240

```
int i = 1;
while (i <= 3)
    System.out.print(i = ++i);
Output: 234
```

Phase 6 – Variable Scope and Lifetime

Q241

```
int a = 5;
{
    int b = 10;
```

```
    System.out.println(a);  
}
```

Output: 5

Q242

```
{  
    int a = 10;  
    System.out.println(a);  
}
```

Output: 10

Q243

```
int a = 5;  
{  
    System.out.println(a);  
}
```

Output: 5

Q244

```
int a = 5;  
{  
    a = 10;  
}  
System.out.println(a);
```

Output: 10

Q245

```
int a = 5;  
{  
    int a = 10;  
    System.out.println(a);  
}
```

```
}
```

Output: **✗ Compilation Error**

Q246

```
int a = 5;
{
    int b = 10;
}
System.out.println(b);
```

Output: **✗ Compilation Error**

Q247

```
int x = 3;
if (x > 0) {
    int y = 5;
    System.out.println(y);
}
```

Output: 5

Q248

```
int x = 3;
if (x > 0) {
    int y = 5;
}
System.out.println(x);
```

Output: 3

Q249

```
for (int i = 0; i < 3; i++) {
    System.out.print(i);
}
```

Output: 012

Q250

```
for (int i = 0; i < 3; i++) {  
}  
System.out.println(i);
```

Output: **✗ Compilation Error**

Q251

```
int a = 5;  
if (a > 0) {  
    int b = a;  
    System.out.println(b);  
}
```

Output: 5

Q252

```
int a = 5;  
if (a > 0) {  
    int a1 = 10;  
    System.out.println(a1);  
}  
System.out.println(a);
```

Output:

10
5

Q253

```
int x = 10;
{
    int x1 = x + 5;
    System.out.println(x1);
}
```

Output: 15

Q254

```
int a = 1;
for (int i = 0; i < 3; i++) {
    a += i;
}
System.out.println(a);
```

Output: 4

Q255

```
int a = 10;
for (int i = 0; i < 2; i++) {
    int b = a++;
}
System.out.println(a);
```

Output: 12

Q256

```
int a = 5;
{
    int b = a++;
    System.out.println(b);
}
System.out.println(a);
```

Output:

5
6

Q257

```
int x = 5;
for (int i = 0; i < 2; i++) {
    x += i;
}
System.out.println(x);
```

Output: 6

Q258

```
int a = 10;
{
    int b = a;
    a = 20;
    System.out.println(b);
}
```

Output: 10

Q259

```
int a = 5;
{
    int b = a;
    {
        int a1 = 10;
        System.out.println(b);
    }
}
```

Output: 5

Q260


```
int a = 1;
for (int i = 0; i < 3; i++) {
    int a1 = i;
}
System.out.println(a1);
```

Output: **✗ Compilation Error**

Q261

```
int a = 5;
if (true) {
    a = 10;
}
if (true) {
    int a1 = 20;
}
System.out.println(a);
```

Output: 10

Q262

```
int x = 1;
for (int i = 0; i < 3; i++) {
    x += i;
}
for (int i = 0; i < 2; i++) {
    x += i;
}
System.out.println(x);
```

Output: 5

Q263

```
int a = 5;
{
    int b = a++;
}
{
```

```
    int b = a++;  
}  
System.out.println(a);
```

Output: 7

Q264

```
int a = 10;  
for (int i = 0; i < 2; i++) {  
    int b = a++;  
    {  
        int c = b + a;  
    }  
}  
System.out.println(a);
```

Output: 12

Q265

```
int a = 1;  
{  
    int b = a;  
}  
{  
    int b = a + 1;  
}  
System.out.println(a);
```

Output: 1

Phase 7 – Array

Q266

```
int[] arr = {1, 2, 3};  
System.out.println(arr[0]);
```

Output: 1

Q267

```
int[] arr = {1, 2, 3};  
System.out.println(arr[2]);
```

Output: 3

Q268

```
int[] arr = {1, 2, 3};  
arr[1] = 5;  
System.out.println(arr[1]);
```

Output: 5

Q269

```
int[] arr = new int[3];  
System.out.println(arr[0]);
```

Output: 0

Q270

```
int[] arr = new int[3];  
arr[0] = 10;  
System.out.println(arr[0]);
```

Output: 10

Q271

```
int[] arr = {1, 2, 3};  
System.out.println(arr.length);
```

Output: 3

Q272

```
int[] arr = {5, 10, 15};  
System.out.println(arr[arr.length - 1]);
```

Output: 15

Q273

```
int[] arr = {1, 2, 3};  
System.out.println(arr[1] + arr[2]);
```

Output: 5

Q274

```
int[] arr = {1, 2, 3};  
arr[0] = arr[2];  
System.out.println(arr[0]);
```

Output: 3

Q275

```
int[] arr = {1, 2, 3};  
System.out.println(arr[1 * 1]);
```

Output: 2

Q276

```
int[] arr = {1, 2, 3};
```

```
for (int i = 0; i < arr.length; i++)  
    System.out.print(arr[i]);
```

Output: 123

Q277

```
int[] arr = {1, 2, 3};  
for (int i = arr.length - 1; i >= 0; i--)  
    System.out.print(arr[i]);
```

Output: 321

Q278

```
int[] arr = {1, 2, 3};  
for (int i = 0; i < arr.length; i++)  
    arr[i] = arr[i] + 1;  
System.out.println(arr[1]);
```

Output: 3

Q279

```
int[] arr = {1, 2, 3};  
int x = arr[0];  
arr[0] = arr[2];  
arr[2] = x;  
System.out.println(arr[0] + arr[2]);
```

Output: 4

Q280

```
int[] arr = {1, 2, 3, 4};  
int sum = 0;  
for (int i = 0; i < arr.length; i++)  
    sum += arr[i];  
System.out.println(sum);
```

Output: 10

Q281

```
int[] arr = {1, 2, 3};
for (int i = 0; i < arr.length; i++)
    arr[i] = i;
System.out.println(arr[2]);
```

Output: 2

Q282

```
int[] arr = {1, 2, 3};
for (int i = 0; i < arr.length - 1; i++)
    arr[i] = arr[i + 1];
System.out.println(arr[0]);
```

Output: 2

Q283

```
int[] arr = {1, 2, 3};
int[] b = arr;
b[0] = 10;
System.out.println(arr[0]);
```

Output: 10

Q284

```
int[] arr = {1, 2, 3};
int count = 0;
for (int i = 0; i < arr.length; i++)
    if (arr[i] > 1)
        count++;
System.out.println(count);
```

Output: 2

Q285

```
int[] arr = {2, 4, 6};
for (int i = 0; i < arr.length; i++)
    arr[i] /= 2;
System.out.println(arr[2]);
```

Output: 3

Q286

```
int[] arr = {1, 2, 3};
for (int i = 0; i < arr.length; i++)
    System.out.print(arr[arr.length - 1 - i]);
```

Output: 321

Q287

```
int[] arr = {1, 2, 3, 4};
for (int i = 1; i < arr.length; i++)
    arr[i] = arr[i] + arr[i - 1];
System.out.println(arr[3]);
```

Output: 10

Q288

```
int[] arr = {1, 2, 3};
int i = 0;
while (i < arr.length)
    System.out.print(arr[i++]);
```

Output: 123

Q289

```
int[] arr = {1, 2, 3};
for (int i = 0; i < arr.length; i++)
    if (i == 1)
        arr[i] = arr[i + 1];
System.out.println(arr[1]);
```

Output: 3

Q290

```
int[] arr = {1, 2, 3};
for (int i = 0; i <= arr.length - 1; i++)
    arr[i] = arr[i] * 2;
System.out.println(arr[2]);
```

Output: 6

Q291

```
int[] arr = {1, 2, 3};
for (int i = 0; i < arr.length; i++)
    arr[i] = arr[(i + 1) % arr.length];
System.out.println(arr[0]);
```

Output: 2

Q292

```
int[] arr = {1, 2, 3, 4};
for (int i = 0; i < arr.length / 2; i++) {
    int temp = arr[i];
    arr[i] = arr[arr.length - 1 - i];
    arr[arr.length - 1 - i] = temp;
}
System.out.println(arr[1]);
```

Output: 3

Q293

```
int[] arr = {1, 2, 3};  
int i = 0;  
arr[i++] = arr[++i];  
System.out.println(arr[0]);
```

Output: 3

Q294

```
int[] arr = {1, 2, 3, 4};  
for (int i = 1; i < arr.length; i++)  
    arr[i] = arr[i] - arr[i - 1];  
System.out.println(arr[3]);
```

Output: 1

Q295

```
int[] arr = {1, 2, 3};  
int[] b = arr;  
arr[1] = 10;  
System.out.println(b[1]);
```

Output: 10

Q296

```
int[] arr = {1, 2, 3};  
for (int i = 0; i < arr.length; i++)  
    arr[i] = arr[i++] + 1;  
System.out.println(arr[1]);
```

Output: 2

Q297

```
int[] arr = {1, 2, 3};  
int sum = 0;
```

```
for (int i = 0; i < arr.length; i++)  
    sum += arr[i++];  
System.out.println(sum);
```

Output: 4

Q298

```
int[] arr = {1, 2, 3};  
int x = arr[0];  
for (int i = 0; i < arr.length - 1; i++)  
    arr[i] = arr[i + 1];  
arr[arr.length - 1] = x;  
System.out.println(arr[2]);
```

Output: 1

Q299

```
int[] arr = {1, 2, 3};  
for (int i = 0; i < arr.length; i++) {  
    int j = i;  
    arr[j] = arr[i] + j;  
}  
System.out.println(arr[2]);
```

Output: 5

Q300

```
int[] arr = {1, 2, 3};  
int i = 0;  
while (i < arr.length)  
    arr[i] = arr[++i];  
System.out.println(arr[0]);
```

Output: Runtime Exception ArrayIndexOutOfBoundsException

Phase 8 – Strings

Q301

```
String s = "A";  
System.out.println(s);
```

Output: A

Q302

```
String s = "A";  
s = s + "B";  
System.out.println(s);
```

Output: AB

Q303

```
String s = "Hello";  
System.out.println(s.length());
```

Output: 5

Q304

```
String s = "ABC";  
System.out.println(s.charAt(1));
```

Output: B

Q305

```
String s = "A";  
System.out.println(s + 1);
```

Output: A1

Q306

```
String s = "10";  
System.out.println(s + 5);
```

Output: 105

Q307

```
String s = "";  
System.out.println(s.length());
```

Output: 0

Q308

```
String s = "Java";  
s = s.concat("Lang");  
System.out.println(s);
```

Output: JavaLang

Q309

```
String s = "ABC";  
System.out.println(s.substring(1));
```

Output: BC

Q310

```
String s = "ABC";  
System.out.println(s.substring(0, 2));
```

Output: AB

Q311

```
String a = "Java";  
String b = "Java";  
System.out.println(a == b);
```

Output: true

Q312

```
String a = new String("Java");  
String b = new String("Java");  
System.out.println(a == b);
```

Output: false

Q313

```
String a = "Java";  
String b = new String("Java");  
System.out.println(a == b);
```

Output: false

Q314

```
String a = "Java";  
String b = a;  
b = b + "Lang";  
System.out.println(a);
```

Output: Java

Q315

```
String a = "A";  
String b = a + "B";  
System.out.println(a == b);
```

Output: false

Q316

```
String a = "A";  
String b = "A" + "B";  
String c = "AB";  
System.out.println(b == c);
```

Output: true

Q317

```
String a = "A";  
String b = a + "B";  
String c = "AB";  
System.out.println(b == c);
```

Output: false

Q318

```
String s = "ABC";  
for (int i = 0; i < 3; i++)  
    s = s + i;  
System.out.println(s);
```

Output: ABC012

Q319

```
String s = "";  
for (int i = 1; i <= 3; i++)  
    s += i;  
System.out.println(s);
```

Output: 123

Q320

```
String s = "A";  
s.concat("B");  
System.out.println(s);
```

Output: A

⚠ immutability trap

Q321

```
String s = null;  
System.out.println(s + "A");
```

Output: nullA

Q322

```
String s = null;  
System.out.println(s);
```

Output: null

Q323

```
String s = "ABC";  
System.out.println(s.indexOf("B"));
```

Output: 1

Q324

```
String s = "ABC";  
System.out.println(s.contains("AB"));
```

Output: true

Q325

```
String s = "ABC";  
System.out.println(s.replace("A", "X"));
```

Output: XBC

Q326

```
String a = "Hello";  
a.concat("World");  
System.out.println(a.length());
```

Output: 5

Q327

```
String a = "Hello";  
String b = a.concat("World");  
System.out.println(a == b);
```

Output: false

Q328

```
String a = "Hi";  
String b = "H" + "i";  
System.out.println(a == b);
```

Output: true

Q329

```
String a = "Hi";  
String b = "H";  
b = b + "i";  
System.out.println(a == b);
```


Output: false

Q330

```
String s = "ABC";  
System.out.println(s.charAt(s.length() - 1));
```

Output: C

Q331

```
String s = "ABCDE";  
System.out.println(s.substring(1, s.length() - 1));
```

Output: BCD

Q332

```
String s = "A";  
for (int i = 0; i < 3; i++)  
    s = s + s;  
System.out.println(s);
```

Output: AAAAAAAAA

Q333

```
String s = "A";  
s = s + 1 + 2;  
System.out.println(s);
```

Output: A12

Q334

```
String s = "A";
```

```
s = 1 + 2 + s;  
System.out.println(s);
```

Output: 3A

Q335

```
String s = "ABC";  
s = s.substring(0, 1) + s.substring(2);  
System.out.println(s);
```

Output: AC

Q336

```
String s = "ABC";  
String t = s;  
s = s.replace("A", "X");  
System.out.println(t);
```

Output: ABC

Q337

```
String s = "ABC";  
String t = s.replace("A", "X");  
System.out.println(s == t);
```

Output: false

Q338

```
String s = "AB";  
s = s + s.length();  
System.out.println(s);
```

Output: AB2

Q339

```
String s = "ABC";  
for (int i = 0; i < s.length(); i++)  
    System.out.print(s.charAt(i));
```

Output: ABC

Q340

```
String s = "ABC";  
for (int i = 0; i < s.length(); i++)  
    s = s + i;  
System.out.println(s);
```

Output: **Infinite loop**

Q341

```
String s = "ABC";  
int i = 0;  
while (i < s.length()) {  
    System.out.print(s.charAt(i));  
    i++;  
}
```

Output: ABC

Q342

```
String s = "A";  
String t = s;  
s += "B";  
t += "C";  
System.out.println(s + t);
```

Output: ABAC

Q343

```
String s = "ABC";  
System.out.println(s.substring(1, 1));
```

Output: `` (empty string)

Q344

```
String s = "ABC";  
System.out.println(s.substring(3));
```

Output: `` (empty string)

Q345

```
String s = "ABC";  
System.out.println(s.substring(0, 0));
```

Output: `` (empty string)

Phase 9 – Type Casting & Promotion

Q346

```
int a = 10;  
double b = a;  
System.out.println(b);
```

Output: 10.0

Q347

```
int a = 5;  
double d = a / 2;  
System.out.println(d);
```

Output: 2.0

Q348

```
double d = 5 / 2;  
System.out.println(d);
```

Output: 2.0

Q349

```
double d = 5.0 / 2;  
System.out.println(d);
```

Output: 2.5

Q350

```
char c = 'A';  
int x = c;  
System.out.println(x);
```

Output: 65

Q351

```
int x = 'a';  
System.out.println(x);
```

Output: 97

Q352

```
byte b = 10;  
int x = b + 5;  
System.out.println(x);
```

Output: 15

Q353

```
float f = 10;  
System.out.println(f);
```

Output: 10.0

Q354

```
int a = 5;  
float f = a;  
System.out.println(f);
```

Output: 5.0

Q355

```
char c = 66;  
System.out.println(c);
```

Output: B

Q 356

```
byte a = 10;  
byte b = 20;  
byte c = a + b;  
System.out.println(c);
```

Output: **✗ Compilation Error**

Q357

```
byte a = 10;  
byte b = 20;  
byte c = (byte)(a + b);  
System.out.println(c);
```

Output: 30

Q358

```
byte b = 10;  
b = b + 1;  
System.out.println(b);
```

Output: **✗ Compilation Error**

Q359

```
byte b = 10;  
b += 1;  
System.out.println(b);
```

Output: 11

Q360

```
int a = 130;  
byte b = (byte)a;  
System.out.println(b);
```

Output: -126

Q361

```
char c = 'A';  
c = c + 1;  
System.out.println(c);
```

Output: **✗ Compilation Error**

Q362

```
char c = 'A';  
c += 1;  
System.out.println(c);
```

Output: B

Q363

```
short s = 10;  
s = s * 2;  
System.out.println(s);
```

Output: **✗ Compilation Error**

Q364

```
short s = 10;  
s *= 2;  
System.out.println(s);
```

Output: 20

Q365

```
int a = 5;  
double d = (double)a / 2;  
System.out.println(d);
```

Output: 2.5

Q366

```
int a = 5;  
double d = a / 2.0;
```



```
System.out.println(d);
```

Output: 2.5

Q367

```
char c = 'a';  
int x = c + 1;  
System.out.println(x);
```

Output: 98

Q368

```
byte b = 100;  
b = (byte)(b * 2);  
System.out.println(b);
```

Output: -56

Q369

```
int a = 5;  
float f = a / 2;  
System.out.println(f);
```

Output: 2.0

Q370

```
float f = 5 / 2;  
System.out.println(f);
```

Output: 2.0

Q371

```
double d = (int)(5.9 + 1.1);  
System.out.println(d);
```

Output: 7.0

Q372

```
int a = (int)5.9 + (int)6.1;  
System.out.println(a);
```

Output: 11

Q373

```
char c = (char)(65 + 2.5);  
System.out.println(c);
```

Output: C

Q374

```
byte b = (byte)300;  
System.out.println(b);
```

Output: 44

Q375

```
int a = 10;  
long l = a * 100000000000L;  
System.out.println(l);  
Output: 1000000000000
```

Phase 10 – Mixed Logic Challenges

Q376

```
int x = 2, y = 4;  
System.out.println(x++ + ++y * x + y++);
```

Output: 22

Q377

```
int a = 1, b = 2;  
System.out.println(++a * ++b + a++ + b++);
```

Output: 11

Q378

```
int x = 5, y = 3;  
System.out.println(x-- * --y + x-- + --y);
```

Output: 15

Q379

```
int a = 3, b = 1;  
System.out.println(a++ + ++b * a + b++);
```

Output: 13

Q380

```
int x = 2, y = 3;  
System.out.println(++x + y++ + x++ * ++y + x);
```

Output: 25

Q381

```
int a = 1;  
if (a++ == 1)  
    System.out.print(a);
```

Output: 2

Q382

```
int a = 5;
for (int i = 0; i < 2; i++)
    a += i;
System.out.println(a);
```

Output: 6

Q383

```
int a = 2;
System.out.println(a++ + ++a);
```

Output: 6

Q384

```
int[] arr = {1, 2, 3};
for (int i = 0; i < arr.length; i++)
    arr[i] += i;
System.out.println(arr[2]);
```

Output: 5

Q385

```
String s = "A";
for (int i = 0; i < 3; i++)
    s += i;
System.out.println(s);
```

Output: A012

Q386

```
int a = 0;
while (a++ < 2)
    System.out.print(a);
```

Output: 12

Q387

```
int a = 5;
if (a > 3 && a++ > 5)
    System.out.print("X");
else
    System.out.print(a);
```

Output: 6

Q388

```
int[] arr = {2, 4, 6};
int sum = 0;
for (int x : arr)
    sum += x;
System.out.println(sum);
```

Output: 12

Q389

```
String a = "Hi";
String b = a;
a += "!";
System.out.println(b);
```

Output: Hi

Q390

```
int a = 3;
if (a-- > 2)
    if (--a > 1)
        System.out.print("A");
```

```
else
    System.out.print("B");
```

Output: B

Q391

```
int i = 0;
for (; i < 3; )
    System.out.print(i++);
```

Output: 012

Q392

```
int[] arr = {1, 2, 3};
int i = 0;
arr[i++] = arr[++i];
System.out.println(arr[0]);
```

Output: 3

Q393

```
String s = "ABC";
System.out.println(s.substring(1) + s.charAt(0));
```

Output: BCA

Q394

```
int a = 1;
do {
    System.out.print(a);
    a++;
} while (a < 1);
```

Output: 1

Q395

```
int a = 5;
for (int i = 0; i < 3; i++)
    if (i == 1)
```

```
        continue;
    else
        a++;
    System.out.println(a);
```

Output: 7

Q396

```
int a = 0;
if (a++ == 0 || ++a == 2)
    System.out.print(a);
```

Output: 1

Q397

```
int[] arr = {1, 2, 3};
for (int i = 1; i < arr.length; i++)
    arr[i] += arr[i - 1];
System.out.println(arr[2]);
```

Output: 6

Q398

```
String s = "A";
s = s + s.length();
System.out.println(s);
```

Output: A1

Q399

```
int i = 3;
while (i-- > 0)
    System.out.print(i);
```

Output: 210

Q400

```
int a = 1;  
System.out.println(a++ + ++a + a);
```

Output: 6

Q401

```
int a = 2;  
if (++a == 3 && a++ == 3)  
    System.out.print(a);  
else  
    System.out.print(a);
```

Output: 4

Q402

```
int a = 1;  
for (int i = 0; i < 3; i++){  
    a = a++ + i;  
}  
System.out.println(a);
```

Output: 4

Q403

```
int[] arr = {1, 2, 3};
```



```
for (int i = 0; i < arr.length; i++)  
arr[i] = arr[i++] + 1;
```

```
System.out.println(arr[1]);
```

Output: 2

Q404

```
String s = "A";  
String t = s;  
for (int i = 0; i < 2; i++)  
    s += i;  
System.out.println(t);
```

Output: A

Q405

```
int i = 0;  
while (i < 3)  
    System.out.print(i = i++ + 1);
```

Output: 123

Q406

```
int a = 3;  
if (a++ == 3)  
    if (++a == 5)  
        System.out.print("X");  
    else  
        System.out.print("Y");
```

Output: X

Q407

```
int[] arr = {1, 2, 3};
```

```
int sum = 0;
for (int i = 0; i < arr.length; )
    sum += arr[i++];
System.out.println(sum);
```

Output: 6

Q408

```
String s = "ABC";
for (int i = 0; i < s.length(); i++)
    s = s + i;
System.out.println(s);
```

Output: Infinite loop

Q409

```
int a = 5;
System.out.println(a > 3 || ++a > 6 && a++ > 5);
```

Output: true

Q410

```
int i = 1;
do {
    System.out.print(i);
    i = i * 2;
} while (i < 3);
```

Output: 12

Q411

```
int a = 1;
System.out.println(a++ + ++a * a++);
```

Output: 9

Q412

```
int[] arr = {1, 2, 3};  
for (int i = 0; i < arr.length - 1; i++)  
    arr[i] = arr[i + 1];  
System.out.println(arr[0]);
```

Output: 2

Q413

```
String s = "AB";  
s = s + s.length() + s.charAt(0);  
System.out.println(s);
```

Output: AB2A

Q414

```
int a = 0;  
while (a++ < 3)  
    if (a == 2)  
        continue;  
    else  
        System.out.print(a);
```

Output: 13

Q415

```
int a = 2;  
System.out.println(a++ + ++a * ++a);
```

Output: 16

Q416

```
int[] arr = {1, 2, 3};  
int x = arr[0];  
arr[0] = arr[2];
```

```
arr[2] = x;  
System.out.println(arr[1]);
```

Output: 2

Q416

```
String s = "A";  
String b = s.concat("B");  
System.out.println(s == b);
```

Output: false

Q417

```
int i = 0;  
for (; i < 3; i++)  
    System.out.print(i = i++);
```

Output: 012

Q418

```
int a = 1;  
System.out.println(++a + a++ + ++a);
```

Output: 7

Q419

```
int a = 3;  
if (a > 2)  
    if (a++ > 3)  
        System.out.print("X");  
    else  
        System.out.print(a);
```

Output: 4

Q420

```
int a = 1;
```

```
System.out.println(a++ + ++a + a++ + ++a);
```

Output: 14

Q421

```
int i = 1;
while (i < 5)
    System.out.print(i = i++ + 1);
```

Output: 2345

Q422

```
int[] arr = {1, 2, 3};
int i = 0;
while (i < arr.length)
    arr[i] = arr[++i];
System.out.println(arr[0]);
```

Output: Runtime Exception

Q423

```
String s = "A";
for (int i = 0; i < 3; i++)
    s = s + s;
System.out.println(s.length());
```

Output: 8

Q424

```
int a = 2;
System.out.println(++a + a++ * a++ + a);
```

Output: 20

Q425

```
int a = 1;
if (a++ == 1 && ++a == 3 || a++ == 3)
    System.out.print(a);
```

Output: 3

Q426

```
int[] arr = {1, 2, 3};
for (int i = 0; i < arr.length; i++)
    arr[i] = arr[(i + 1) % arr.length];
System.out.println(arr[0]);
```

Output: 2

Q427

```
String s = "ABC";
String t = s;
s = s.replace("A", "X");
System.out.println(t);
```

Output: ABC

Q428

```
int a = 3;
System.out.println(a++ + ++a * a + a++);
```

Output: 19

Q429

```
int i = 0;
do {
    System.out.print(i);
    i = i++ + 1;
} while (i < 3);
```

Output: 012

Q430

```
int a = 5;
for (int i = 0; i < 2; i++)
    a = a++ + ++a;
System.out.println(a);
```

Output: 26

Q431

```
String s = "A";  
String b = s + "B";  
String c = "AB";  
System.out.println(b == c);
```

Output: false

Q432

```
int[] arr = {1, 2, 3};  
int sum = 0;  
for (int x : arr)  
    sum += x++;  
System.out.println(sum);
```

Output: 6

Q433

```
int a = 0;  
while (++a < 3)  
    System.out.print(a++);
```

Output: 1

Q434

```
int a = 1;  
System.out.println(a++ + ++a * ++a + a++);
```

Output: 16

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